

A Jai-inspired systems language in Rust. Incremental salsa front end, lossless CST, typed SSA mid-end, two execution engines held byte-identical by a differential harness.

6/6 gates green 919 tests ADR-0062 latest W3 in progress · 14 waves uncommitted

| TESTS                  | CORPUS                 | ADRS                    | DIAGNOSTICS            | EDITOR CHECKS              |
|------------------------|------------------------|-------------------------|------------------------|----------------------------|
| 919                    | 144                    | 62                      | 88                     | 151                        |
| workspace, all passing | jr files, both engines | 0001 to 0062, immutable | codes, E0258 next free | Neovim, verified not gated |

The vertical slice, and what it costs to be honest about it

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| <p>EXIT CRITERIA</p> <p><b>DONE</b> hello.jr runs in the VM</p> <p><b>DONE</b> compiles to a native binary</p> <p><b>DONE</b> rustc-grade diagnostics</p> <p><b>DONE</b> formatter round-trips the corpus</p> <p><b>DONE</b> editor integration (Neovim)</p> <p><b>OPEN</b> verified Linux x86-64 CI run</p> <p>The last one needs a push. Configured, never run, so it is a decision rather than a technical gap.</p> | <p>HOW CORRECTNESS IS ESTABLISHED</p> <p><b>Both engines run every corpus program</b> and must agree, on output and on trap wording, from one shared formatter.</p> <p><b>MIR snapshots pin the exact instruction sequence</b> for all 138 files. Last wave they changed for every file that calls anything; this wave they are what shows a #no_abc procedure lowering without the check its neighbour keeps.</p> <p><b>Every fix is verified by reverting it</b> and watching the test fail. A test that passes without its fix is worse than none — and this wave two grammar rules were reverted to prove their checks had teeth.</p> <p><b>The gates cannot see everything.</b> context is a legal identifier, so a tree-sitter grammar that had never heard of it still parsed the corpus cleanly while meaning something else. Gate 6 was green throughout. That class of drift is now checked on the node type instead.</p> |
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The language today

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| <p>WORKS, END TO END</p> <p>Full integer tower, bool, string, pointers</p> <p>float32 and float64, plain IEEE-754, no traps</p> <p>struct, nominal, one level</p> <p>union, nominal, untagged, fields at offset 0</p> <p>enum and enum_flags, namespaced, bare dot-member</p> <p>Fixed arrays and views, bounds-checked, returnable</p> <p>cast and xx from context; operator overloading</p> <p>Trapping arithmetic, wrapping variants, bitwise</p> <p>if, else, while, for, break, continue, defer, using</p> <p>Multiple returns, named args, literal defaults</p> <p>import, foreign, system_library, #scope_module</p> <p>One trivial compile-time run, folded</p> <p>Traps name their source line</p> <p>Bounds checks, strippable by build setting or #no_abc</p> <p>context, a hidden parameter passed by pointer</p> <p>Procedures as values: call, pass, return, struct field</p> <p>null, a context-typed pointer literal; malloc and free</p> <p>An allocator in the context, installed and called through</p> | <p>ABSENT (WAVE)</p> <p>pointer arithmetic, deliberately none</p> <p>percent on floats, is_nan, math (W7)</p> <p>a tagged variant type (W4.5)</p> <p>explicit backing type; a switch (W4.5)</p> <p>array literals, sub-slicing</p> <p>unary, index and call overloading</p> <p>transmute; float printing</p> <p>#must; a multi-result call in a return</p> <p>polymorphs, macros (W5)</p> <p>arbitrary run, RTTI, insert (W4)</p> <p>backtraces (W3)</p> <p>a per-index #no_abc; any other build setting</p> <p>allocators, temp storage (W3)</p> <p>a cross-file or foreign proc value; comparing one</p> <p>pointer arithmetic; cast to a pointer</p> <p>push_context; a bump allocator; temp storage (W3)</p> |
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No error-handling model yet — ADR-0008 reserves the slot and the first half exists (several return values); #must is owed its own ADR. No GC and no RAI, which is a design value rather than a gap. A union is untagged, so reading a field other than the one last written reinterprets bits: documented in three places because it cannot be diagnosed. No indirect calls, which is the single largest gap in the project and is what blocks the rest of W3.

Compiler internals, 17 crates

| STAGE                             | STATE       | NOTE                                                             |
|-----------------------------------|-------------|------------------------------------------------------------------|
| Lexer, parser, CST, typed AST     | works       | Hand-written, error-recovering, trivia-preserving                |
| Formatter                         | works       | Pure function over the CST; lost source in 7 of the last 8 waves |
| HIR, name resolution, modules     | works       | Flat import merge; cycles legal; export filtering                |
| InternPool: types, values, layout | works       | One layout computation and one integer evaluator, shared         |
| Sema: signatures, checking        | works       | E0212 to E0257; no const-eval here, by design                    |
| MIR: typed SSA                    | works       | Block parameters, notphis; explicit bounds check and zeroing     |
| Mid-end                           | 5 passes    | Inline, forwarding, const-prop, DCE, plus the bounds-check strip |
| Const-eval                        | works       | Runs MIR through the bytecode VM                                 |
| VM: register bytecode, libffi     | works       | No JIT; indirect calls, and malloc from its own region           |
| Cranelfit back end                | works       | Aggregate returns via sret; indirect calls via func_addr         |
| LLVM back end                     | not started | W8 owns it                                                       |
| Language server                   | 12 caps     | Diagnostics, hover, goto, completion, rename, actions, hints     |
| Neovim integration                | works       | Runtimepath dir, no plugin manager; 151 checks                   |
| Driver                            | stub        | Should consume the workspace notion that now exists              |

Roadmap: W1 and W2 closed, W3 started

|                        |             |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W1 Data                | done        | Numeric tower, wrapping ops, enum, enum_flags, union, arrays, views, cast, xx, operator overloading. Closed by ADR-0048.                                                                                                                                                                                                                                                                                          |
| W2 Flow and scope      | done        | for with it and it_index, labelled break, defer, using, multiple returns, named and default arguments, #scope_module. Closed by ADR-0054; ADR-0055 then closed a gap six corpus files had carried for eleven waves.                                                                                                                                                                                               |
| W3 Runtime core        | in progress | Five of six done. context (ADR-0057), the bounds-check build setting (ADR-0058, finishing ADR-0003), indirect calls (ADR-0059), null plus a memory source (ADR-0060/0061), and the allocator protocol (ADR-0062) — a program installs an allocator in the context and a callee allocates through it without knowing which. Remaining: temporary storage (which wants push_context too) and traps with backtraces. |
| W4 Comptime            | not started | Arbitrary compile-time execution, RTTI and Type values, insert, code. PLAN section 5 names this the project's top risk: sema and comptime become mutually recursive.                                                                                                                                                                                                                                              |
| W4.5 Pattern matching  | not started | switch with exhaustiveness, a bare dot-member as a case, and a tagged variant type. Was missing from the wave table entirely — two accepted ADRs deferred decisions to it while no wave scheduled it.                                                                                                                                                                                                             |
| W5 Polymorphism        | not started | Polymorphic parameters, modify, bake_arguments, expand macros with hygiene, instantiation caching.                                                                                                                                                                                                                                                                                                                |
| W6 Metaprogram         | not started | Workspaces, the compiler message loop, build scripts replacing makefiles, note attributes.                                                                                                                                                                                                                                                                                                                        |
| W7 Stdlib              | not started | Written in Jairs: String, dynamic array, hash table, Sort, Math, Random, File.                                                                                                                                                                                                                                                                                                                                    |
| W8 Performance         | not started | LLVM via inkwell for release builds, inliner maturity, struct-of-arrays, SIMD, parallel sema. Also owns the compile-throughput number, still unpublished.                                                                                                                                                                                                                                                         |
| W9 Tooling depth       | not started | Semantic tokens are the one LSP capability still missing; the other twelve landed early.                                                                                                                                                                                                                                                                                                                          |
| W10 Graphics, in Jairs | not started | Window creation through foreign calls, Metal then Vulkan, immediate-mode 2D.                                                                                                                                                                                                                                                                                                                                      |

What this session did, and how each wave was scoped

**FOURTEEN WAVES SHIPPED**

ADR-0049 through 0062: for and defer, using, aggregate returns, multiple returns, named and default arguments, scope visibility, imported constants, float constants, context, the bounds-check build setting, indirect calls, null plus a memory source, and the allocator protocol.

Test count 900 to 919. Corpus 116 to 144 files. Neovim checks 103 to 151. W2 closed; W3 opened and five of its six features landed. A program can now install an allocator in the context and a callee allocates through it without knowing which.

**Each wave was scoped by writing the feature and seeing what the compiler refused**, not by reading the handoff. That found gaps the handoff had missed in both directions: a proc-pointer struct field it called absent (it worked), and a void-returning proc-pointer type it never mentioned (unspellable, and blocking).

**TWO CLAIMS THE CODE DISPROVED**

**An ADR corrected mid-wave.** ADR-0060 §4 asserted the VM dereferences a malloc'd pointer via libffi. Running the corpus file the same ADR promised would pass disproved it: the VM's memory is a linear region and a host address is not an offset into it. ADR-0061 corrects it — the VM allocates from its own region, native calls libc, the bits differ and nothing observes them.

**A diagnostic that could not be acted on.** Installing an imported #foreign procedure into an allocator field reported “expected (s64) -> \*u8, found (s64) -> \*u8” — the same text twice, because the types differ only in an invisible calling convention. It is E0256 now, the code that says to wrap it.

**THE THING WORTH YOUR DECISION**

Fourteen waves sit uncommitted across five branches. Two waves ago a careless git checkout reverted the tree-sitter grammar nine waves and cost an hour's reconstruction — the concrete price of not committing. Committing each wave as it goes green would bound the damage from any future slip to a single wave.

*Recorded as a recommendation, not a change: the authorisation is yours to give.*

Sources: PLAN.md §1.5 and §7, the ADR directory, docs/decisions/DECISIONS.md, and the six gates run today. Every number was measured rather than carried forward — the test count from a full workspace run, the corpus count from a file walk, the diagnostic codes from the const E0nnn definitions across the crates, and the editor-check count from a real verify.lua run. Rebuild with typst compile jairs-dashboard.typ.